

Developer Manual — COMP3211 PIM

Group: 89. Member names and student IDs: pending human input.

Product baseline: frozen source, tests, and coverage tool at commit 4d3508db0447a3ac6346a2a2789da89208f88fd5 (2026-09-30). This manual documents that build. **Platform:** Windows 11 with PowerShell. **Verified Python versions:** 3.13.5 and 3.12.3. Python 3.11 is a requirements target but was not installed in the WP04 environment and remains unverified. No third-party runtime or test package is needed.

1. Open a clean checkout

Use Git with access to the group's private repository. These commands place the shell in the repository root and select the exact frozen product build:

```
git clone https://github.com/hzihao872-arch/COMP3211-PIM.git
Set-Location COMP3211-PIM
git switch --detach 4d3508db0447a3ac6346a2a2789da89208f88fd5
git rev-parse HEAD
python --version
```

The `git rev-parse HEAD` output must be the full freeze SHA above. The `python --version` output must identify the interpreter selected for the following commands. WP04 executed the application and suite with Python 3.13.5 and 3.12.3 on Windows 11. The release documents may be on a later commit; checking out the freeze SHA is for reproducing the product evidence. The same commands also run from a later documentation checkout while `src/`, `tests/`, and `tools/model_coverage.py` still match that SHA.

For a clean checkout already open at the repository root, start with `git rev-parse HEAD` and verify the freeze SHA before reproducing WP04 exactly. A Python virtual environment is optional; all product imports are from the standard library or this repository.

2. IDE guidance and project layout

Visual Studio Code with a Python extension is a suitable editor on Windows 11; no specific IDE version was part of WP04 verification. Open the **repository root folder**, select the installed Python 3.13.5 or 3.12.3 interpreter, and use its integrated PowerShell terminal for the commands below. The terminal commands are authoritative for reproduction; an IDE-specific test or coverage display was not used to obtain the report.

Path	Responsibility
<code>src/main.py</code>	Composition root; creates a <code>PIMManager</code> and starts <code>CommandController</code> .
<code>src/model/records.py</code>	Immutable Note, Task, Event, and Contact records.
<code>src/model/validation.py</code> , <code>errors.py</code>	Model value rules and domain errors.
<code>src/model/manager.py</code>	Current collection, IDs, create/get/list/update/delete/search, validated snapshot reconstruction.
<code>src/model/search.py</code>	Search atoms, Boolean criteria, tokenizer, parser, and evaluator.
<code>src/controller/cli.py</code>	Command tokenization, dispatch, formatting, and recoverable error messages.
<code>src/storage/pim_file.py</code>	Version 1 UTF-8 JSON .pim save/load and validation.
<code>tests/model/</code> , <code>tests/controller/</code> , <code>tests/storage/</code> , <code>tests/test_acceptance.py</code>	Model unit, controller, persistence, and real-process acceptance tests.
<code>tools/model_coverage.py</code>	Standard-library trace runner for the model line-coverage report.

The model package has no dependency on controller or storage. The controller imports the model and storage boundary. `src/main.py` is the only process entry point.

3. Run, test, and inspect coverage

Execute each block from the repository root. The program launches with an empty in-memory collection:

```
python src/main.py
```

At the `pim>` prompt, type `help` for the full command grammar, then `exit` to end the session. A short smoke session is `help`, `add note "memo"`, `show 1`, `exit`. Data is not saved automatically; use `save "records.pim"` and later `load "records.pim"` when persistence is wanted.

Run the automatically discovered suite:

```
$env:PYTHONPATH = "$PWD\src"
python -m unittest discover -s tests -t . -v
```

The explicit `-t .` prevents `tests/model` from shadowing `src/model`. WP04 observed **33 tests, zero failures, zero errors** on both verified interpreters. Its Python 3.12.3 run selected that interpreter with `py -3.12 -m unittest discover -s tests -t . -q` after setting the same `PYTHONPATH`. The local fresh-clone check at the freeze SHA also ran the CLI and 33 tests successfully. Test elapsed time varies by machine.

Generate the submitted model-only line-coverage numbers:

```
python tools/model_coverage.py
```

This tool discovers and runs the same 33 tests under Python's standard-library `trace` module. The WP04 Python 3.13.5 report was **411/430 executable model lines (95.58%)**. It covers `src/model/*.py`; subprocess executions of `src/main.py` in acceptance tests are outside the trace scope. The exact per-module figures and method appear in the separate Test Coverage Report and `docs/reports/model-coverage-raw.txt`.

Check that source, tests, and the coverage tool compile:

```
python -m compileall -q src tests tools
```

Successful `compileall` produces no output and exits with code 0. It creates ignored bytecode cache folders.

4. Debug a command

Set the IDE's working directory to the repository root, program to `src/main.py`, interpreter to the verified Python selected above, and use an integrated terminal so you can type CLI commands. No command-line arguments are required. If the IDE needs an import path for direct test debugging, set `PYTHONPATH` to the repository's `src` folder; launching `src/main.py` directly already puts `src` on Python's import path.

Useful breakpoints in the frozen code are `src/main.py::main`, `src/controller/cli.py::CommandController._dispatch` and `.run`, `src/model/manager.py::PIMManager.update` OR `.search`, `src/model/search.py::parse_criterion`, and `src/storage/pim_file.py::save_pim` OR `load_pim`. For a failed search, follow `parse_criterion` into `_Parser.parse`, then inspect `SearchSyntaxError` handling in `CommandController.run`. For a failed load, inspect `load_pim` before the controller replaces its active manager. The controller prints expected failures with `Error: SEARCH, NOT_FOUND, VALIDATION, PATH, FORMAT, IO, or COMMAND` and returns to the next prompt.

5. Evidence boundary

The course's NFR-02 includes Python 3.11. `py -3.11 --version` found no suitable runtime during WP04 and the WP05 clean-checkout check, so this manual does not claim Python 3.11 compatibility has been verified. IDE guidance is a setup recommendation; WP04 evidence came from PowerShell commands. Group 89 is confirmed, while member identity fields remain for human completion.